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Research Paper

Does FDI increase product innovation of domestic firms? Evidence from China ☆

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ABSTRACT

Exploiting a change in policy governing the entry of foreign direct investment (FDI) in 2002, we apply the difference-in-differences model to estimate the effects of FDI on the product scope of domestic Chinese firms. In industries that experienced relaxation in FDI regulations, the average product scope increased by 5% which indicates more product innovation. Through vertical linkages, the product scope of firms is positively (negatively) affected by FDI in upstream (downstream) industries. The negative effect of FDI in downstream industries occurs because foreign firms engaging in processing trade rely more on imported inputs than inputs from domestic suppliers. The main channels of effect are firm-level R&D and industry-level technological distance, as FDI entry leads to an improvement in these variables. However, the positive effects of FDI on product innovation have limits, as new products are mainly introduced within the same industry rather than across different industries.

1. Introduction

Foreign direct investment (FDI) plays a crucial role in the global economy by facilitating the transfer of physical capital, technology, managerial practices, and high-skilled workers across national borders. A significant body of literature has been dedicated to addressing the fundamental question of whether FDI generates positive spillover effects in the host country. Besides the influence on productivity and employment, the effect of FDI on innovation is of particular interest. Although it is well understood that product innovation is a crucial aspect of economic growth and development (Kuznets, 1973; Romer, 1990), the analysis of the effects of FDI

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on product innovation is unfortunately constrained by the available measures of product innovation. Only a few studies (Kee, 2015; Javorcik et al., 2018) directly examine the change in the mix of products produced by firms, but the available evidence remains scant, with the estimated effects ranging from insignificant to positive.

As a high-growth economy with prominent status in both global FDI and the global manufacturing landscape, the Chinese economy presents an excellent context for studying the dynamic relations between FDI and product innovation. Starting from zero foreign investment at the beginning of the era of ‘reform and opening up’ in 1978, China has actively courted FDI. By the end of 2022, China boasted the second largest stock of inward FDI, surpassed only by the United States.¹ Simultaneously, China has become one of the largest contributors to world manufacturing, accounting for approximately 31% of world manufacturing value-added in 2022. As China’s manufacturing sector grows, it demonstrates a nontrivial capacity for product upgrades and innovation.

It is crucial to inquire whether the vast amount of FDI that China has managed to attract contributes to product innovation of Chinese firms. The answer to this question is significant for understanding innovation in China and other developing countries which are interested in attracting FDI. To address this question, our study utilizes the China Product Output Database, a large and unique firm-level data set that, to the best of our knowledge, has not been analyzed in the literature before. Leveraging the change in policy governing FDI entry in 2002 that accompanied China’s WTO entry in 2001, we apply the difference-in-differences (DID) estimation strategy to identify the causal effects.

Our analysis of Chinese firms during the period of 2000–2007 yields four results on product innovation. Firstly, FDI enhances the product scope of domestic firms in the same industry. In industries that experienced relaxation in FDI entry immediately after China’s entry into the WTO, the product scope of domestic firms increases by 5% on average. Secondly, FDI entry also affects product scope through vertical linkages. The product scope of domestic firms is positively related to FDI in upstream industries but negatively related to FDI in downstream industries. The positive effect of upstream FDI aligns with the theory and evidence in the literature which suggest that foreign firms upstream can exert a positive influence by offering superior or more cost-effective inputs. We show that the negative effect of downstream FDI occurs mainly in industries characterized by a higher prevalence of processing trade, a trade regime that weakens the link between foreign firms and domestic suppliers. Thirdly, there is some evidence suggesting that a firm-level channel of FDI effects is R&D, as firm-level R&D expenditures increase by 5% following the change in FDI entry policy. In terms of the industry-level channel, FDI entry leads to a shortening of the technological distance between domestic industry and the world frontier. Fourthly, the positive effects of FDI have limits. New products are primarily added in the same industry rather than across different industries, indicating that most product innovation is incremental rather than radical.

The positive effects of FDI on product scope are robust when considering three factors that potentially contribute to expanding the product scope—trade liberalization, imported inputs, and reduction in trade policy uncertainty. Additionally, when utilizing alternative measures of product innovation, such as the rate of new product addition and the probability of product innovation, the positive effects of FDI persist. To further strengthen our analysis, we employ alternative identification methods, including the event study method and synthetic difference-in-differences (SDID). The estimated effects of FDI on product scope, based on the alternative methods, remain statistically and practically significant.

This study is closely related to Kee (2015) and Javorcik et al. (2018), which both examine the relationship between FDI and product scope. Kee (2015) demonstrates that FDI increases the product scope in Bangladesh firms and that an important effect channel is common local suppliers shared by both foreign and domestic firms. As foreign firms demand better input from local suppliers, the improvement in quality also benefits domestic firms that use the same suppliers. In contrast, Javorcik et al. (2018) does not find a significant connection between FDI and the introduction of new products by domestic firms in Turkey but finds an increase in the complexity of their products. In the context of China, Ito et al. (2012) and Li et al. (2013) find positive effects of FDI on two indicators closely related to product innovation, patent innovation and the share of new products in sales.

Our findings are relevant to the broad literature on the spillover effects of FDI through both horizontal and vertical linkages. Theoretically, FDI can generate positive spillovers in the same industry, known as horizontal linkages. This is because foreign firms often demonstrate better technology to domestic firms and train workers who later take up jobs in domestic firms (Teece, 1977; Blomström and Kokko, 1998). Additionally, FDI improves peer domestic firms’ access to better inputs as foreign firms help improve the technological capacity of local suppliers (Kee, 2015). However, foreign firms naturally want to guard their technology, which mutes spillovers. Moreover, an increase in competition can negatively affect the performance of domestic firms. In empirical studies, the productivity spillovers from foreign to domestic firms are mixed, ranging from positive in Mexico (Kokko, 1994), the US (Keller and Yeaple, 2009), and Bangladesh (Kee, 2015), to insignificant in Turkey (Javorcik, 2004), and to negative in Venezuela (Aitken and Harrison, 1999).

Regarding vertical channels, the existing literature posits positive forward spillovers in general (Blomström and Kokko, 1998) but empirical evidence remains limited (Gorodnichenko et al., 2014). In the forward channel, the introduction of better or cheaper inputs by foreign firms in upstream industries increases product innovation among domestic firms.² As for backward linkage, the effects of foreign firms in downstream industries depend on the nature of the linkage (Rodríguez-Clare, 1996). If foreign firms help expand the market for domestic inputs or voluntarily transfer technology to domestic firms upstream, positive spillover through backward linkages is expected. Conversely, if foreign firms displace domestic inputs with imported inputs, negative spillovers may occur. Most empirical studies find that productivity spillovers through backward linkages are positive (Javorcik, 2004; Blalock and Veloso, 2007;

¹ The ranking is based on the *World Investment Report 2023* published by the United Nations Conference on Trade and Development.

² Goldberg et al. (2010) show that imported inputs made available by trade liberalization lead to more new products introduced by local firms in India, which can be viewed as indirect evidence for positive forward spillovers. This is because both FDI entry in upstream industries and trade liberalization improve the availability of better inputs.

Girma and Gong, 2008). However, Xu and Sheng (2012) report negative productivity spillovers through backward linkages in the context of China.

Relative to existing studies, we make two contributions. First, using the unique data in the China Product Output Database, this study is the first to estimate the effect of FDI on Chinese firms' product scope. The results reveal that, in general, FDI plays a significant role in contributing to product innovation among Chinese firms. Our finding complements existing studies on FDI spillovers in China that have predominantly focused on outcomes such as productivity. However, the product innovation associated with FDI entry tends to be incremental rather than radical.

Second, we provide a comprehensive set of results that examine the effects of FDI on product innovation through both the horizontal and vertical linkages. The horizontal spillovers are positive as FDI increases the product scope of domestic firms in the same sector. Regarding vertical linkages, the product scope of domestic Chinese firms is positively related to the entry of FDI in upstream industries but negatively related to FDI in downstream industries. The negative backward spillovers contrast with the findings from other countries, albeit with the caveat that these studies focus on productivity. Through further analysis, we uncover that the negative effect through the backward channel occurs because entry of foreign firms in industries with prevalent processing trade weakens the link between these industries and their domestic suppliers. Overall, our study contributes to the understanding of the complexity of FDI spillovers by utilizing a rich set of information from multiple micro datasets specifically gathered from China.

The remainder of this paper is organized as follows. We present the empirical research design and describe the data in Section 2 and report the empirical results regarding the effects of FDI on product scope in Section 3. In Section 4, we explore the channels of the FDI effects. The robustness of the results and issues of heterogeneity are discussed in Section 5. Finally, in Section 6, we conclude the paper.

2. Empirical design

2.1. Difference-in-difference estimation

To estimate the effect of FDI entry on product scope, we adopt the DID regression of the product scope of a firm on a policy change that affects FDI entry in some industries in 2002. The regression equation is specified as follows:

$$\ln(scope_{fit}) = \beta_0 + \beta_1 treat_i \times post_t + \rho X_{fit} + \gamma_f + \delta_t + \epsilon_{fit} \quad (1)$$

where f , i , and t are the indices for the firm, the 4-digit industry, and the year, respectively. Following Kee (2015), our dependent variable is the log of a firm's product scope ($\ln(scope)$), which is measured as the number of products produced by firm f in year t in industry i . The variable $treat_i$ is a dummy variable that equals 1 if industry i becomes more open to FDI entry in 2002 and 0 if the industry experienced no change in entry policy. The group of industries that experienced no policy changes was the reference group. The variable $post_t$ is an indicator, which takes the value of 1 for years after 2002 and 0 otherwise. The parameter of the interaction term, β_1 , captures the effect of the relaxation of FDI entry on a firm's product scope. If $\beta_1 > 0$, then the data support the notion that the policy change in 2002 promotes the development of product scope. Quantities γ_f and δ_t are firm fixed effects and time fixed effects, respectively.

Our control variables are the following: firm size, measured as log output ($\ln Output$); log firm age ($\ln Age$); capital intensity, measured as log capital to labor ratio ($\ln KL$); financing ability ($FinAbility$); an indicator for state-owned enterprises (SOE); an indicator for exporting activities ($export$). The error term was ϵ_{fit} . We clustered the standard error at 4-digit industry level.

2.2. Overhaul of FDI policy in 2002

In this subsection, we discuss the overhaul of FDI policy in the revision of the Catalogue of Industries for Guiding Foreign Investment (hereinafter, the Catalogue of Industries) in 2002. Published by the State Council of China, the Catalogue of Industries plays a fundamental role in the approval process of FDI projects by indicating the industries in which FDI is encouraged.³ Over time, the revisions to the Catalogue of Industries indicate how China has become increasingly open to FDI.

The 2002 revision of the Catalogue of Industries constitutes a substantial relaxation in FDI entry regulations in China. By announcing that FDI was encouraged in a larger number of industries, the 2002 Catalogue of Industries significantly increased the attractiveness of these industries to foreign investors. Crucially, the 2002 revision was part of China's efforts to fulfill its obligations associated with entry into the WTO in 2001.⁴ As part of the WTO entry arrangement, China agreed to increase its economic openness significantly and substantially reduce the barrier to inward FDI. The timing of WTO entry and subsequent policy changes was determined by a long and difficult negotiation process punctuated by geopolitical factors and other shocks and hence arguably exogenous to foreign investors. In particular, the revision of the Catalogue of Industries was unrelated to the contemporaneous level of product scope in China.

³ The Catalogue of Industries was first published in 1995 and underwent revisions in 1997, 2002, 2004, 2007, 2011, 2015, and 2017. In 2019, 2020 and 2022, the State Council further consolidated the Catalogue and released the Catalogue of Industries for Encouraging Foreign Direct Investment.

⁴ During our sample period of 2000 to 2007, China made changes to the Catalogue of Industries in 2002, 2004, and 2007. Of these, the revision in 2002 was the most substantial because it was linked to the obligations associated with China's WTO entry. The revision to the Catalogue of Industries in 2004 was minor, serving mainly to fine-tune industrial policies. Thus, we choose the revision in 2002 as the main policy shock in our sample period and study the effects of the resulting FDI inflow on the product scope of firms.

In the Catalogue of Industries, there is a classification of industries into four categories in descending order of FDI entry: encouragement, permission, hindrance, and forbiddance. In the first step of identifying and coding changes in FDI policy, we compare the Catalogue of Industries of 1997 to that of 2002 and record the change (or the lack of change) in status of an industry's FDI entry policy as encouraged, unchanged, or tightened. Thus, we identify the changes in FDI policies for a large number of industries.

In the second step, to link the changes in FDI policy to other data, we map the manufacturing industries in the Catalogue of Industries to 424 manufacturing industries in the Chinese Industry Classification (CIC) 4-digit industry codes.⁵ As the former classification is finer than the latter, our mapping procedure sometimes places two or more industries in the former classification into a single 'umbrella' industry in the latter. However, multiple industries from the former classification that go into the same CIC 4-digit 'umbrella' industry could experience different policy changes. Recognizing this complexity, we distinguish three cases. First, if industries from the Catalogue of Industries experienced either relaxation or no change in entry policy, we coded the CIC 4-digit 'umbrella' industry as 'encouraged'. Second, if industries from the Catalogue of Industries experienced either tightening or no change in entry policy, we code the CIC 4-digit 'umbrella' industry as 'tightened'. Finally, if among the industries from the Catalogue of Industries, at least one experienced relaxation and at least one experienced tightening in entry policy, we code the CIC 4-digit 'umbrella' industry as 'mixed'.

After applying the above two-step procedure, among the 424 CIC 4-digit industries, we code 106 industries as 'encouraged',⁶ 294 as 'unchanged', 13 as 'tightened', and 11 as 'mixed'. In the DID estimation, we defined the 'encouraged' industries as the treated group and the unchanged industries as the control group.⁷ While it is plausible that the revision of the Catalogue of Industries in 2002 was an exogenous policy event, some industry characteristics likely affect whether a particular industry experiences a relaxation in FDI regulation. To address the potential endogeneity problem associated with differences in pre-existing trends, we control for several industry characteristics in 2000 that are related to innovation and productivity. They are the fraction of new products in sales, export-to-sales ratio, average wage rate, and the number of firms in the industry.⁸

2.3. Data and measurement

Our sample covers all large and medium-sized manufacturing firms in China from 2000 to 2007. The two main data sources are the *Annual Survey of Industrial Firms* (ASIF) and the *China Product Output Database* (CPOD), both obtained from the National Bureau of Statistics of China (NBSC). ASIF, widely used in research on Chinese firms, covers private firms with sales greater than 5 million RMB (approximately US 760,000 at the market exchange rate) and all SOEs. On average, there were approximately 230,000 unique firm observations each year, from 2000 to 2007. The survey provides rich information on two sets of variables: those related to basic information and those related to financial information. Variables related to basic information include identification code, firm name, industry, and geographical location. Important examples of financial variables are output value, sales, exports, employment, and sources of paid-in capital (state, foreign, and domestic private).

We obtained product scope information from CPOD data, which covers more than 200,000 industrial firms per year in China from 2000 to 2007. For each firm, the database provides information on the number of products produced, the 5-digit product codes, product names, and quantities. A product is identified using a 5-digit product code. As ASIF and CPOD share the same firm identification codes issued by the NBSC, linking the two databases is straightforward. The merged CPOD and ASIF data accounted for 90.90% of the CPOD and 37.60% of the ASIF.⁹

Following standard practices in the field (Chor et al., 2021), we exclude observations unusable or unreliable for the following reasons: (1) reporting zero or negative firm age, industrial output, sales, fixed assets, and industrial value added; (2) employing fewer than eight employees; (3) reporting negative export value or an export value greater than sales. After these adjustments, we were left with 615,416 firm-year observations covering the period from 2000 to 2007.

We exclude foreign firms to examine the spillover effect of FDI entry on domestic firms. We define a firm as foreign if foreign capital (including capital from Hong Kong, Macau, and Taiwan) accounts for more than 25% of the paid-in capital. Subsequently, to avoid the extreme values problem, we winsorized the continuous variable at the 1st and 99th percentiles.

We define product scope as the number of 5-digit CPOD product codes produced by a firm. Our practice is similar to that of Bernard et al. (2010), which uses 5-digit SIC codes to identify products. In the Appendix, we show that the 5-digit CPOD codes we

⁵ We use the version of CIC codes developed by Brandt et al. (2012), as the authors provide consistent concordance between different vintages of industry classification published by the National Bureau of Statistics of China.

⁶ These 106 4-digit industries belong to 23 2-digit manufacturing industries, which account for the majority of 29 of China's 2-digit manufacturing industries. They include low-, medium-, and high-tech industries. The six industries that did not experience relaxation in entry, with their 2-digit codes in parentheses, are the following: the manufacturing industry of textile costumes, shoes, and caps (18), wood processing and manufacturing industry of wood, bamboo, rattan, palm, and straw-made articles, (20), furniture (21), printing and reproduction of record media (23), plastic product (30), and ferrous metal smelting and extrusion (32).

⁷ The results remain essentially unchanged if we use industries coded as 'unchanged', 'tightened', and 'mixed' as the control group.

⁸ If the selection of industries into the control and treatment groups is not random—the two groups of industries have pre-existing differences in the tendency to experience relaxation in FDI regulation before the revision of the Catalogue of Industries in 2002—our DID estimation will be inconsistent. Presumably, the government is likely to ease or retain FDI regulation in selected industries to suit the development goals. For instance, the government may want to limit high-pollution industries and promote high-tech industries. To account for the non-randomness in selection into treatment, we follow Lu et al. (2017) to control for industry characteristics that likely affect FDI regulation.

⁹ Changes in FDI regulations, as explained in Section 2.2, are identified at the level of 4-digit industries. As the industry classification in the ASIF data changed from GB/T4754-1994 (2000–2002) to GB/T4754—2002 (2003–2007) in 2003, we ensured consistency of the industry classification by applying the concordance table developed by Brandt et al. (2012).

Table 1
Summary Statistics

Type of firm	Share of firms average	Average product scope	Share of output	Share of employment	Average TFP
Single-product firms	67.83%	1	40.03%	45.29%	10.03
Multi-product firms	32.17%	2.51	59.97%	54.71%	10.16
	Mean	Observations	Standard Deviation	Min	Max
scope	1.42	338,201	0.73	1	5
lnScope	0.26	338,201	0.40	0	1.61
treat	0.36	338,201	0.48	0.00	1
post	0.79	338,201	0.41	0	1
lnOutput	10.16	338,201	1.21	6.62	14.03
lnAge	2.18	338,201	0.90	0	4.03
lnKL	0.04	338,201	0.01	0.004	0.06
FinAbility	0.05	338,201	0.08	-0.02	0.74
export	0.20	338,201	0.40	0	1
SOE	0.12	338,201	0.33	0	1

Note: This table reports the number of observations, mean, standard deviation, minimum and maximum value for the following key variables: firm product scope and its log, the fraction of observations subject to treatment, the fraction of observations in the post policy change period, log of firm output, log of firm age, log of firm capital-intensity, financing ability, indicator for exporter, and indicator for SOEs.

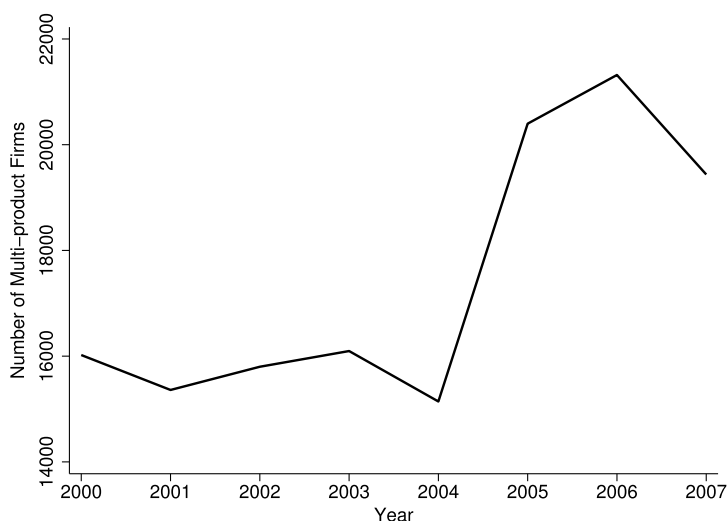


Fig. 1. Numbers of Multi-Product Firms

Source: authors' calculation based on the CIPO and ASIF.

use can be mapped into the 5-digit codes in the Central Product Codes (CPC version 1.0) published by the United Nations. Thus, the 5-digit CPOD codes can accurately identify products.

As shown in Table 1, 67.83% of firm-year observations correspond to single-product firms. For firm-year observations reporting multiple products, the average number of products is 2.51. Fig. 1 shows that the number of multi-product firms increased during our sample period. Fig. 2 shows that multi-product firms are superior to their single-product counterparts, reporting, on average, higher output, larger sales value, more employment, and higher total factor productivity.

As for measures of control variables, we use the logged value of a firm's output as a proxy for firm size (*lnOutput*). Our measure of a firm's age (*lnAge*) is the log of a firm's total number of years in business plus one. We use the interest expenses to fixed capital ratio to measure the financing ability of a firm. This is because firms subject to fewer financial constraints can borrow more funds externally and incur a larger amount of interest rate expenses against the same level of fixed capital. The larger this indicator (*FinAbility*), the less financially constrained is the firm. We define a firm as an SOE if the state accounts for more than 50% of its paid-in capital. Finally, the export indicator (*export*) is equal to 1 if a firm registers positive exports in a given year and 0 otherwise.

3. Main regression results

3.1. Baseline results

We report the DID regression results associated with equation (1) in Table 2. In the first column, we include the treatment variable (*treat* $\times$ *post*), predetermined industry-level controls, year fixed effects, and firm fixed effects. We then introduce firm-level control variables in the regression in column (2). In column (3) of Table 2, which is our preferred benchmark, we control for two concurrent

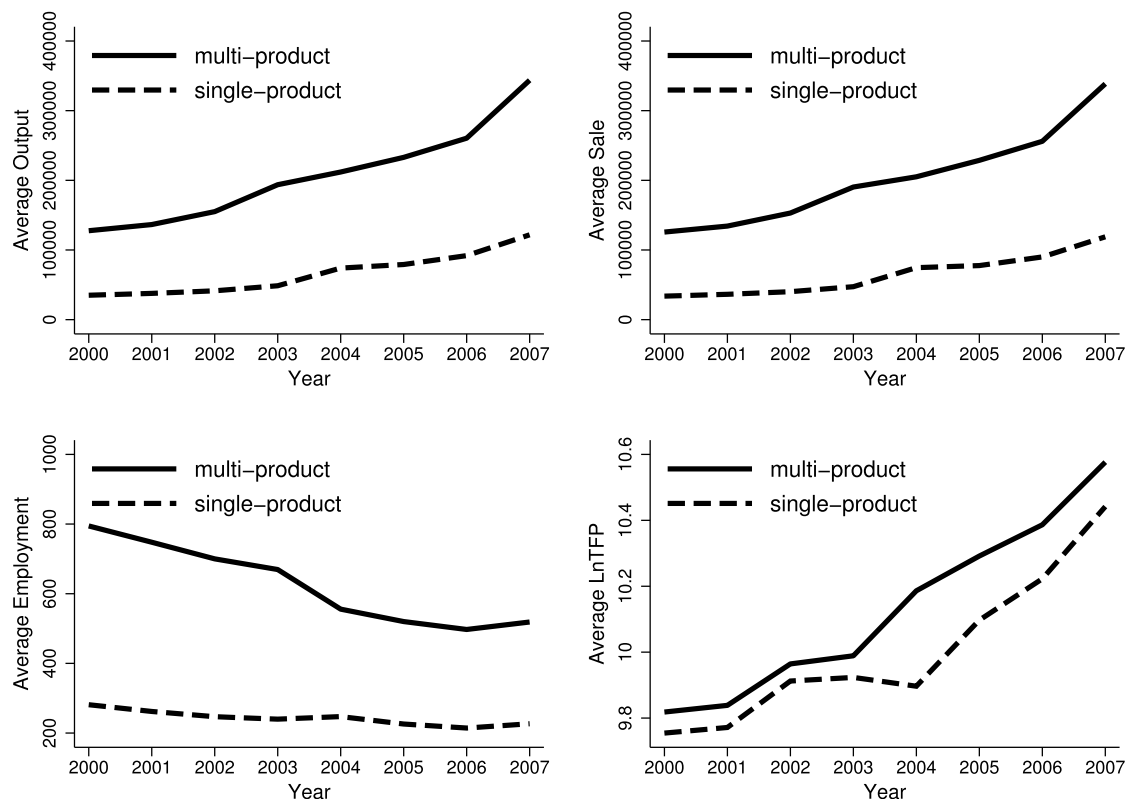


Fig. 2. Performance Differences between Multi-product Firms and Single-product Firms

Source: authors' calculation based on the CIPO and ASIF.

Table 2

Baseline Results for FDI Spillovers through the Horizontal Linkage

Dependent Variable: lnScope	(1)	(2)	(3)
treat × post	0.06** (0.03)	0.06** (0.03)	0.05*** (0.02)
lnOutput		0.01*** (0.002)	0.01*** (0.002)
lnAge		0.004 (0.003)	0.01** (0.002)
lnKL		0.02 (0.19)	0.06 (0.16)
FinAbility		-0.001 (0.01)	0.002 (0.01)
export		0.01*** (0.004)	0.01*** (0.004)
SOE		0.01*** (0.005)	0.01** (0.005)
Dependent Variable's Mean	0.26	0.26	0.26
Firm fixed effects	Y	Y	Y
Year fixed effects	Y	Y	Y
Pre-determined industry controls	Y	Y	Y
Tariff in 2001×year dummy	N	N	Y
Proportion of SOE in 2001×year dummy	N	N	Y
Constant	0.44*** (0.11)	0.29** (0.12)	0.21** (0.08)
Observations	338,201	338,201	338,201
R-squared	0.89	0.89	0.89

Notes: 1) The variables *treat*, *lnOutput*, *lnAge*, *lnKL*, *FinAbility*, *export*, and *SOE* are the treatment variable associated with the 2002 policy change, log of firm output, log of firm age, log of firm's capital intensity, firm's financial ability, indicator for exporter, and indicator for SOE. Tariff is the 4-digit industry-level ad valorem tariff in 2001.

2) ***, **, * denote significance at the 1%, 5% and 10% level, respectively. The standard errors reported in parentheses are clustered at the level 4-digit industry.

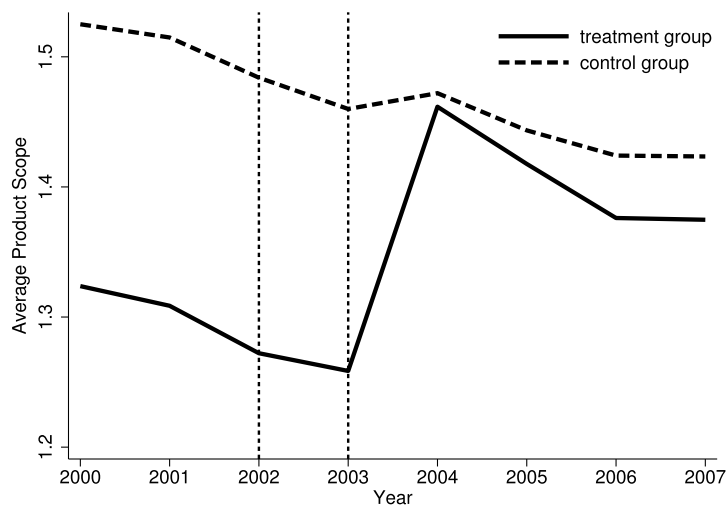


Fig. 3. Trends of Average Product Scope for Treatment Group and Control Group

Source: authors' calculation based on the CIPO and ASIF.

reforms: China's entry into the WTO in 2001 and the reform of SOEs. China's entry into the WTO ushered in an era of further opening up of the country, which could have affected the product scope through trade liberalization. With the creation of the China State-owned Assets Supervision and Administration Commission in 2003, existing SOEs underwent significant reforms that allowed them far greater flexibility in deciding the product mix and product scope. To account for the potential effects of these reforms, we include in the regressions the interactions of year dummies and the import tariff of 2001 and those of year dummies and the fraction of SOEs in the number of firms in 4-digit industries in 2001. The estimated effects are significant in all three specifications.

The estimated coefficient of the treatment variable is 0.05 in the benchmark regression in Column (3). The estimate indicates that firms in the treated industries—industries that experienced deregulation in FDI entry—report a growth in product scope that is 5% ($0.05 \times 100\%$) greater than that of firms in industries in the control group. Compared to the mean of the log product scope (0.26), the effect of FDI entry is also significant in the economic sense.

3.2. Parallel trend test

When using the DID to evaluate the effect of a policy change, the control and treatment groups must satisfy the parallel trend assumption. Before the policy change, we need to ensure that the two groups have similar trends; hence, pre-existing differences do not confound the effect of the policy. In Fig. 3, we plot the trends in product scope for the control group (firms in industries that experienced no relaxation in FDI entry policy) and the treatment group (firms in industries that experienced relaxation in FDI entry). As shown in Fig. 3, the trends for the two groups were similar prior to 2003, and the treatment group showed a notable increase in product scope relative to the control group. Evidence suggests that the parallel trend assumption is likely to be satisfied in our application.

3.3. Testing for omitted variables bias

In addition to the parallel trend test, we also follow Oster (2019) to test for presence of omitted variables bias that affect selection of industries into treatment and control groups. Because researchers often use additional control variables to reduce the threat to identification caused by omitted variables, Oster (2019) designs a method to use the change in goodness of fit (R^2) associated with the additional controls to detect omitted variable bias. As mentioned in the section of empirical design, the additional controls used in this paper are four predetermined variables, i.e., the fraction of new products in sales, export-to-sales ratio, average wage rate, and the number of firms in the industry. We carry out two tests of omitted variable bias suggested by Oster (2019).

In the first test of omitted variable bias, we construct bounds for treatment effect at the 95% level, based on the assumptions that the degree of correlation between the additional controls with the treatment variable is equivalent to that of the omitted variables (the coefficient of proportionality δ is 1 in terminology of Oster (2019)), and that inclusion of the omitted variables would boost the R^2 to 1.3 times of the current level. As shown in the first column of Table 3, the bounds are [0.046, 0.053]. Because zero does not fall in the interval, it is unlikely that the true treatment effect is zero or negative.

In the second test, we again assume that inclusion of the omitted variables would boost the R^2 to 1.3 times of the current level, and compute the δ value that would force the true treatment effect to zero. As shown in column (2) of Table 3, the critical δ value is 2.16. The value means that if the degree of correlation between the omitted variables with the treatment variable is 2.16 times that of the additional controls, then the treatment effect is zero. Because this would imply a very high level of relative explanatory power of omitted variables (well above the critical value of $\delta = 1$ suggested by Oster (2019)), again it is unlikely that true treatment effect is zero or negative. Therefore, our results pass the two commonly-used tests of omitted variable bias.

Table 3
Test of Omitted Variables Bias Based on Oster (2019)

Test	(1) bounds of treatment effects	(2) coefficient of proportionality (δ)
Assumptions	$R_{max}^2 = 1.3R^2$, $\delta = 1$	$R_{max}^2 = 1.3R^2$, true treatment effect = 0
Criterion	0 is excluded from the interval	$\delta > 1$
Estimated values	[0.046, 0.053]	$\delta = 2.16$
Pass or not	yes	yes

Notes: 1) R_{max}^2 denotes the hypothetical R^2 that could be obtained if the omitted variables are included in the regression.

2) The coefficient of proportionality δ measures the degree of correlation between the additional controls with the treatment variable relative to that of the omitted variables.

3) The bounds of treatment effects are constructed for the 95% confidence level.

3.4. Vertical linkages

In addition to the horizontal spillover effects of FDI, we consider spillovers that occur along the supply chain or vertical linkage. To capture the effect of FDI on product scope along the vertical linkage, we follow Javorcik (2004) to construct the following forward FDI linkage measure:

$$treat_i^{forward} \times post_t = \sum_{s \neq i} \alpha_{si} \times treat_s \times post_t$$

$$\alpha_{si} = \frac{input_{si}}{\sum_s input_{si}},$$

where $input_{si}$ is the input from industry s used in industry i , and α_{si} is the share of industry s in cost of industry i . Similarly, the measure of industry i 's backward FDI linkage is defined as

$$treat_i^{backward} \times post_t = \sum_{k \neq i} \alpha_{ik} \times treat_k \times post_t$$

$$\alpha_{ik} = \frac{output_{ik}}{\sum_k output_{ik}},$$

where $output_{ik}$ is the input from industry i used in industry k , and α_{ik} is the share of industry k in the sales output of industry i . Input and output data were obtained from the 2002 Chinese Input-Output (IO) table.

In Table 4, we add the measures of forward FDI linkage ($treat_i^{forward} \times post_t$) and backward FDI linkage ($treat_i^{backward} \times post_t$) to the benchmark regression first separately and then jointly. The results show that the horizontal effect of FDI on product scope remains significant, and its magnitude is similar to the benchmark. As for vertical linkage, the coefficient of the forward FDI linkage is positive, indicating that the product scope of a firm is positively associated with the entry of FDI in upstream industries. The finding of positive forward linkages is consistent with the theory of Rodriguez-Clare (1996), as domestic firms' innovation can be boosted by better inputs provided by foreign firms upstream. Our finding of positive spillovers on product scope is also consistent with the previous empirical studies (Ito et al., 2012; Xu and Sheng, 2012), albeit their focus is on productivity spillovers.

Because the weighted sum of indicators for relaxation of FDI regulation in 106 CIC 4-digit industries is used to construct treatment variables for forward and backward linkages, we need to interpret the treatment effects relative to the variation in the respective treatment variables. The means of the horizontal treatment variable ($treat_i \times post_t$), the forward treatment variable ($treat_i^{forward} \times post_t$) and the backward treatment variable ($treat_i^{backward} \times post_t$) are 0.28, 0.09, and 0.08, respectively. The corresponding standard deviations are 0.449, 0.083, and 0.096. Thus the effect of a one-standard-deviation horizontal FDI treatment on log product scope is $0.06 \times 0.449 = 0.027$ and the effect is equivalent to 10.36% of the mean log product scope ($(0.06 \times 0.449 / 0.26) \times 100\% = 10.36\%$). Through similar calculation, the effect of a one-standard-deviation forward (backward, respectively) FDI treatment on log product scope is 0.017 (-0.017) and the effect is equivalent to 6.38% (6.65%) of the mean log product scope. Therefore, adjusted for the variation of the treatment variables, the effects of FDI through vertical linkages are slightly larger than half of the effect of FDI through the horizontal linkage, indicating that they are important in understanding the spillover effects of FDI.

The backward FDI linkage is estimated to exert a negative effect, which is contrary to findings based on data from Lithuania (Javorcik, 2004), Indonesia (Blalock and Veloso, 2007), and the UK (Girma and Gong, 2008) but is consistent with Xu and Sheng (2012) in the context of China. Notably, our outcome variable is the product scope, while the aforementioned set of papers examines productivity. While domestic firms in theory can gain technology transferred by foreign firms downstream, Rodriguez-Clare (1996) notes that if foreign firms downstream prefer imported inputs to domestic inputs, they can have negative effects on domestic suppliers. During our sample period, the prevalence of processing trade is a prominent feature of China's trade which weakens the link between the downstream industries and their domestic suppliers. To be specific, processing trade accounted for 47.2% of the total trade of China in the period. Within processing trade, foreign firms dominated with a share of 81.6%. Therefore, it is highly likely that these foreign firms rely heavily on imported inputs and source less frequently from domestic suppliers. Therefore, we conjecture that entry of foreign firms in downstream industries with significant processing trade negatively affects product innovation of domestic suppliers.

Table 4
FDI Spillovers Along the Vertical Linkages

Dependent Variable: lnScope	(1)	(2)	(3)	(4)
treat _{post}	0.06*** (0.02)	0.05*** (0.02)	0.06*** (0.02)	0.05*** (0.02)
treat ^{forward} _{post}	0.17** (0.08)		0.20** (0.08)	0.22*** (0.08)
treat ^{backward} _{post}		-0.16*** (0.06)	-0.18*** (0.06)	
treat ^{backward} _{low processing} _{post}				-0.05 (0.13)
treat ^{backward} _{high processing} _{post}				-0.24*** (0.08)
Dependent Variable Mean's	0.26	0.26	0.26	0.26
Firm fixed effects	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y
Pre-determined industry controls	Y	Y	Y	Y
Time-varying firm controls	Y	Y	Y	Y
Tariff in 2001×year dummy	Y	Y	Y	Y
Proportion of SOE in 2001×year dummy	Y	Y	Y	Y
Constant	0.20** (0.08)	0.21*** (0.08)	0.20** (0.08)	0.20** (0.08)
Observations	336,825	336,825	336,825	336,825
R-squared	0.89	0.89	0.89	0.89

Notes: 1) The variable *treat* is the treatment variable associated with the 2002 policy change. *treat^{forward}*, *treat^{backward}*, *treat^{backward}_{low processing}* and *treat^{backward}_{high processing}* are treatment variables for forward FDI linkage, backward FDI linkage, backward FDI linkage from industries with below median processing trade, and backward FDI linkage from industries with above median processing trade.

2) The time-varying firm control variables are log of firm output, log of firm age, log of firm's capital intensity, firm's financial ability, indicator for exporter, and indicator for SOE. Tariff is the 4-digit industry-level ad valorem tariff in 2001. Proportion of SOE is calculated by the ratio of SOE in each 4-digit industry in 2001.

3) ***, **, * denote significance at the 1%, 5% and 10% level, respectively. The standard errors reported in parentheses are clustered at the level 4-digit industry.

To test whether foreign firms engaging in processing trade depress product innovation through the backward linkage, we create the following treatment variables for backward linkage from industries with above median processing trade and backward linkage from industries with below median processing trade:

$$treat_{i,high\ processing}^{backward} \times post_t = \sum_{k \neq i} \alpha_{ik} \times treat_k \times 1(high\ processing = 1) \times post_t$$

$$treat_{i,low\ processing}^{backward} \times post_t = \sum_{k \neq i} \alpha_{ik} \times treat_k \times 1(high\ processing = 0) \times post_t,$$

where $1(high\ processing = 1)$ is an indicator function which takes value of 1 if industry k is an industry with a high level of processing trade. Following Liu and Qiu (2016), we define an industry as a high-processing-trade one if the fraction of processing trade in total trade of the industry is above the fraction of processing trade in total trade of all industries in 2000. In column (4) of Table 4, the coefficient on the backward linkage from industries with below median processing trade is insignificant while the coefficient on backward linkage from industries with above median processing trade is negative and significant.¹⁰ Thus, consistent with our conjecture, the negative effects from backward linkage are mainly associated with industries with a high level or processing trade.

4. Analysis of channels

4.1. Firm-level channels

We study two firm-level channels for product scope change: firms' R&D and managerial efficiency. First, we examine whether FDI entry prompts domestic firms to increase their innovation efforts through R&D. To test this channel, we regress the log of the R&D expenditures of local firms on the treatment variable in column (1) of Table 5. The estimates indicate that post treatment, firms in the treated industries experience a 5% increase in R&D expenditure. As the p-value of the coefficient is 0.064, the results provide reasonable support to the R&D channel.

¹⁰ Yu (2015) measures the prevalence of processing trade by the fraction of firms engaged in processing trade in an industry. In unreported regression, we use this measure and find the coefficient on backward linkage from industries with above median processing trade to be -0.28 and statistically significant, which is similar to the results reported in column (4).

Table 5
Firm-level Channels

Dependent Variables:	lnR&D (1)	ME distance (2)
<i>treat</i> × <i>post</i>	0.05* (0.02)	-0.08 (0.06)
Dependent Variable's Mean	0.66	-0.01
Firm fixed effects	Y	Y
Year fixed effects	Y	Y
Pre-determined industry controls	Y	Y
Time-varying firm controls	Y	Y
Tariff in 2001×year dummy	Y	Y
Proportion of SOE in 2001×year dummy	Y	Y
Constant	-1.37*** (0.20)	6.21*** (0.23)
Observations	235,990	261,718
R-squared	0.71	0.34

Note: 1) The dependent variables are log R&D expenditure at firm level, and the distance between a firm's managerial efficiency and the frontier level, respectively.

2) The variable *treat* is the treatment variable associated with the 2002 policy change. The time-varying firm control variables are log of firm output, log of firm age, log of firm's capital intensity, firm's financial ability, indicator for exporter, and indicator for SOE. Tariff is the 4-digit industry-level ad valorem tariff in 2001. Proportion of SOE is calculated by the ratio of SOE in each 4-digit industry in 2001.

3) ***, **, * denote significance at the 1%, 5% and 10% level, respectively. The standard errors reported in parentheses are clustered at the level 4-digit industry.

The second firm-level channel that we consider is managerial efficiency. When foreign firms bring advanced management practices with them, domestic firms can improve their management practices and efficiency by mimicking and learning from foreign firms. Nocke and Yeaple (2006) show that the marginal cost of a firm is determined by the managerial efficiency and optimal product scope, and higher efficiency in management enables a firm to overcome a higher marginal cost and increase its product scope. Using firm-level data from China between 2000 and 2006, Qiu and Yu (2020) find that, in response to the foreign country's tariff cut, firms with better managerial efficiency could expand the product scope of exporters, while firms with lower managerial efficiency experienced reductions in product scope.

To test whether FDI entry is associated with higher managerial efficiency, we use distance to frontier managerial efficiency as the dependent variable in the DID estimation.¹¹ In the second column of Table 5, we report the regression with distance to frontier managerial efficiency as the dependent variable. The coefficient of the treatment variable is negative but not statistically significant, with a p-value of 0.160. Thus, evidence on managerial efficiency as an effect channel is marginal.

4.2. Industry-level channel

We consider two industry-level channels, technological distance and resource misallocation. First, FDI entry can help local firms introduce new products by reducing the industry-level technological distance between China and the world frontier. This is because FDI entry generates technology spillovers through the demonstration effect and human capital training. After receiving the de facto technology transfer, existing firms can choose to upgrade the technology used in current products and start manufacturing new products. To capture the technological distance channel, we modified the benchmark regression model by replacing *lnscope* with technological distance as the dependent variable. The distance is measured as the industry-specific labor productivity ratio, $dist_{it} = \frac{LP_{US_{it}}}{LP_{CN_{it}}}$, where $LP_{CN_{it}}$ and $LP_{US_{it}}$ are the labor productivity of industry *i* in China and the US, respectively. Here, we follow Aghion et al. (2009) by assuming that US industries are at the world technology frontier. The larger this ratio, the larger the technological distance between China and the world frontier.

Column (1) of Table 6 reports the regression of technological distance. The results confirm our conjecture that FDI deregulation reduces the technological distance between a firm's industries and the world frontier. The coefficient on *treat*×*post* is -2.13, implying that FDI deregulation is associated with a 2.13 drop in technological distance. Relative to the mean technological distance of 19.58, this effect represents a sizable improvement of technology at the industry level.

¹¹ Following Qiu and Yu (2020), we measure managerial efficiency as the residual of the following regression of overhead expenses: $lnManage_{f,t} = \beta_1 \times lnL_{f,t} + \beta_2 \times lnExp_{f,t} + \beta_3 \times markup_{f,t} + \gamma_f + \gamma_t + \epsilon_{f,t}$, where $lnManage_{f,t}$ is the log of overhead expenses, $lnL_{f,t}$ is the log of employees, $lnExp_{f,t}$ is the log of firm exports, and $markup_{f,t}$ is the firm's markups. Quantities γ_f and γ_t are the firm fixed effects and the year fixed effects, respectively. The larger the overhead residual for firm *f* in this regression, the lower the firm's managerial efficiency. Subsequently, we define the mean value of the managerial efficiency of firms with managerial efficiency in the top 10% as frontier managerial efficiency. Dividing a firm's managerial efficiency by frontier value, we obtain the measure of distance to frontier managerial efficiency.

Table 6
Industry-level Channels

Dependent Variables:	Technological distance		Misallocation
	All (1)	Tech-density (2)	
$treat \times post$	-2.13*** (0.81)	-1.82* (0.93)	0.01 (0.03)
$treat \times post \times high\text{-}tech$		0.36 (2.27)	
$treat \times post \times low\text{-}tech$		-1.12 (2.20)	
$post \times high\text{-}tech$		-1.03 (1.61)	
$post \times low\text{-}tech$		4.34*** (0.91)	
Dependent Variable's Mean	19.58	19.58	0.96
Industry fixed effects	Y	Y	Y
Year fixed effects	Y	Y	Y
Pre-determined industry controls	Y	Y	Y
Time-varying industry controls	Y	Y	Y
Tariff in 2001 $\times$ year dummy	Y	Y	Y
Proportion of SOE in 2001 $\times$ year dummy	Y	Y	Y
Constant	43.41*** (5.23)	42.57*** (5.14)	-0.005 (0.25)
Observations	3,035	3,035	2,907
R-squared	0.85	0.86	0.47

Notes: 1) The dependent variables are the distance between an industry's labor productivity and the world frontier level, and the degree of resource misallocation at industry level which is measured by the standard deviation of log TFP within an industry, respectively.

2) The variable *treat* is the treatment variable associated with the 2002 policy change. The time-varying industry control variables are log of average industry output, log of average industry age, log of industry's average capital intensity, industry's average financial ability, share of exporter and SOE. Tariff is the 4-digit industry-level ad volorem tariff in 2001. Proportion of SOE is calculated by the ratio of SOE in each 4-digit industry in 2001.

3) ***, **, * denote significance at the 1%, 5% and 10% level, respectively. The standard errors reported in parentheses are clustered at the level 4-digit industry.

In Column (2), we examine whether the effects of FDI on technological distance vary with industry-level technology intensity. As industries with high levels of technology intensity are R&D intensive, it is likely to be more difficult and costly for domestic firms to catch with the world frontier and expand their product scope by learning from foreign firms (Lall, 2000; Manova and Zhang, 2009). We follow Sandven et al. (2005) and Lu et al. (2010) in using the definitions of industry-level technology intensity from the Organization of Economic Cooperation and Development (OECD) to define three types of industries: high-tech, medium-tech, and low-tech industries. A detailed classification is presented in Table A1 in the Appendix. We introduce to channel regression the triple interaction terms involving an indicator for high-tech industries and an indicator for low-tech industries, and the corresponding double interaction terms. To be specified, the regression

$$\begin{aligned}
 tech\ dist_{it} = & \beta_0 + \beta_1 treat_i \times post_t + \beta_2 treat_i \times post_t \times hightech_i + \beta_3 treat_i \times post_t \times lowtech_i \\
 & + \beta_4 post_t \times hightech_i + \beta_5 post_t \times lowtech_i \\
 & + \rho \cdot X_{it} + \gamma_i + \delta_t + \epsilon_{it}
 \end{aligned} \tag{2}$$

In Column (2) of Table 6, the reference group is medium-tech industries and the coefficient on the double interaction term, $treat \times post$, is negative, indicating that entry of FDI reduces technological distance for medium-tech industries. Meanwhile, the triple interaction terms involving high-tech industries and low-tech industries, $treat \times post \times hightech$ and $treat \times post \times lowtech$, are both insignificant. While the overall effect of FDI on technological distance of high-tech industries (low-tech industries, respectively) which equals the sum of coefficients on $treat \times post$ and $treat \times post \times hightech$ ($treat \times post \times lowtech$) is negative, a test of the significance indicates that the effect is not statistically significant.¹² Overall, FDI entry appears to shorten technological distance in industries of three level of technology intensity, however the effect is not statistically significant for high-tech and low-tech industries.

The second industry-level channel is a potential reduction in resource misallocation, which is often considered an important aspect of the Chinese economy. If the entry of foreign firms improves resource allocation within an industry, then domestic firms are better positioned to introduce new products. We use the standard measure of resource misallocation introduced by Hsieh and Klenow

¹² For high-tech industries, the effect of FDI on technological distance is -1.46, which is the sum of the coefficients on $treat \times post$ (-1.82) and $treat \times post \times hightech$ (0.36). The p-value for two-sided test is 0.47. For the low-tech industries, the effect of FDI on technological distance is -2.94, which is the sum of the coefficients on $treat \times post$ (-1.82) and $treat \times post \times lowtech$ (-1.12). While the effect is large in magnitude, it is statistically insignificant with a p-value of 0.15.

Table 7
Heterogeneity: Technology Intensity, Ownership and Upstreamness

Dependent Variable: lnScope	Tech-intensity (1)	Ownership (2)	Upstreamness (3)
<i>treat</i> × <i>post</i>	0.09*** (0.02)	0.05*** (0.02)	0.13*** (0.02)
<i>treat</i> × <i>post</i> × <i>high</i>	0.04 (0.05)		
<i>treat</i> × <i>post</i> × <i>low</i>	-0.09*** (0.03)		
<i>post</i> × <i>high</i>	0.07*** (0.02)		
<i>post</i> × <i>low</i>	-0.01 (0.01)		
<i>treat</i> × <i>post</i> × <i>SOE</i>		-0.004 (0.01)	
<i>treat</i> × <i>SOE</i>		-0.03** (0.02)	
<i>post</i> × <i>SOE</i>		-0.02*** (0.01)	
<i>treat</i> × <i>post</i> × <i>upstream</i>			-0.13*** (0.02)
<i>treat</i> × <i>post</i> × <i>midstream</i>			-0.12*** (0.02)
<i>post</i> × <i>upstream</i>			-0.05*** (0.01)
<i>post</i> × <i>midstream</i>			-0.03** (0.01)
Dependent Variable's Mean	0.26	0.26	0.26
Firm fixed effects	Y	Y	Y
Year fixed effects	Y	Y	Y
Pre-determined industry controls	Y	Y	Y
Time-varying firm controls	Y	Y	Y
Tariff in 2001×year dummy	Y	Y	Y
Proportion of SOE in 2001×year dummy	Y	Y	Y
Constant	0.21*** (0.08)	0.21*** (0.08)	0.19** (0.08)
Observations	338,201	338,201	338,201
R-squared	0.89	0.89	0.89

Notes: 1) The variable *treat* is the treatment variable associated with the 2002 policy change. The time-varying firm control variables are log of firm output, log of firm age, log of firm's capital intensity, firm's financial ability, indicator for exporter, and indicator for SOE. Tariff is the 4-digit industry-level ad valorem tariff in 2001. Proportion of SOE is calculated by the ratio of SOE in each 4-digit industry in 2001.

2) ***, **, * denote significance at the 1%, 5% and 10% level, respectively. The standard errors reported in parentheses are clustered at the level 4-digit industry.

(2009)—the standard deviation of log TFP within an industry—in place of *lnscope* as the dependent variable. However, the results reported in column (3) of Table 6 suggest that FDI entry has no effect on the degree of resource misallocation within an industry.

5. Heterogeneity and robustness results

5.1. Heterogeneity analysis

First, we check whether the effects of FDI deregulation on product scope vary with technology intensity. We introduce to the benchmark product scope regression the triple interaction terms involving indicators for high-tech and low-tech industries, and the corresponding double interaction terms. The lineup of regressors is similar to equation (2). The coefficient on *treat* × *post* is 0.09 in Column (1) of Table 7, indicating that relaxation of FDI regulation is associated with a 9% increase in product scope. Because the triple interaction term including the indicator for high-tech (low-tech, respectively) industries is positive and insignificant (negative and significant), the effect of FDI policy on product in high-tech (low-tech) industries is similar than (significantly smaller than) that of medium-tech industries. We then test the significance of the sum of coefficient on each triple interaction term and the coefficient *treat* × *post* to assess the overall effect of FDI on product scope of high-tech and low-tech industries, respectively. We find that the overall effect on high-tech industries (0.13) is positive and significant at 1% level. Meanwhile, the overall effect on low-tech industries (0.00) is insignificant. Overall, FDI deregulation boosts product scope in high-tech and medium-tech industries, but not in low-tech industries.

In the second heterogeneity analysis, we consider whether the effect of FDI entry on product scope differs between SOEs and domestic private firms. In our sample, the mean product scopes of these two types of firms were 1.80, and 1.42, respectively. The

product scope of SOEs was notably larger, presumably because they had better access to resources, and their product choices were heavily influenced by the state. To estimate whether the effect of FDI deregulation on product scope differs by ownership type, we add the triple interaction term with the SOE indicator ($treat \times post \times SOE$) and the corresponding double interaction term to the benchmark regression. In Column (2) of Table 7, the coefficient of the triple interaction term (-0.004) is small in magnitude and highly insignificant. Thus, there is no statistical difference between the effect of FDI deregulation on product scope of SOEs and the effect on private enterprises.

The third heterogeneity is the upstreamness of a firm's industry. In studies of global value chains, a notable observation is that the upstream stages of production and downstream stages generate more value added than the midstream stages Mudambi (2008). Thus, we conjecture that FDI may motivate more domestic innovation in upstream and downstream industries, in which the potential return on innovation is higher. We adopt the upstreamness concept of Antràs et al. (2012), in which an industry with a larger average distance from final use is considered to be upstream. After sorting the upstreamness level of 71 manufacturing industries based on the 2002 Input-Output table of China, we define the top third of the industries as upstream, the next third as midstream, and the bottom third as downstream. The classification details are presented in Table A2 in the Appendix.

Similar to the previous two heterogeneity analysis, we introduce to the regression the triple interaction terms for the upstream industries and midstream industries, and the corresponding double interaction terms. Based on the results in Column (3) of Table 7, the coefficients on the two triple interaction terms are both negative and significant. Because in term of the magnitude, the coefficient on either of the triple interaction terms almost negates the baseline positive effect measured by coefficient on $treat \times post$, the effects of FDI on product scope of upstream and midstream industries are close to zero in practical sense. Formal statistical tests confirm that these effects are statistically insignificant too. The results suggest that in the period of 2000 to 2007, Chinese firms were not capable of expanding product scope in the upstream and midstream stages of production.

In the final heterogeneity analysis, we distinguish between entrant, incumbent, and exit firms. Following Dunne et al. (1988), for each year t , we define firms that appear in the sample in year t but not in the previous period as entrant firms and firms present in year t but not in the next period as exit firms. The remaining firms are incumbents in year t . In Table 8, we present three regressions to demonstrate the potential heterogeneity. In the first column, we introduce triple interaction terms with entrant indicator and exit indicator and the corresponding double interaction terms, leaving the incumbent as the reference group. The results indicate that FDI entry has a positive effect on product scope of incumbent firms. The effects are weaker for entrants and exit firms but the differences are not statistical significant. In the second column, we remove the intercept and introduce triple interaction terms for all three types of firms to obtain directly the significance level of the FDI effect for each type of firm. The coefficients on entrants, incumbent firms, and exit firms are 0.06, 0.07 and 0.03, respectively, and the last effect is insignificant. Therefore, the effect of FDI on product scope is weaker on exit firms relative to entrant firms and incumbent firms. This is reasonable because firms on the borderline of exiting are less likely to expand product scope. Finally, to exclude the effects of entry and exit dynamics, in column (3) of Table 8, we retain only firms that operate throughout the sample period of 2000–2007. The estimated effect of FDI deregulation (0.10) is substantially larger than the benchmark effect estimated for the full sample (0.05).

5.2. Alternative dependent variables

First, we analyze the radicality of innovation at the firm level by examining whether a firm adds a new product to the current industry, a different industry, or even a new sector. Here, we define a sector as a 2-digit CIC code and maintain the definition of an industry as a 4-digit CIC code. In the whole sample, 3.58% of the firm-year observations firms report the addition of products. Within this total, in 2.30% of observations, firms add products in a 4-digit CIC industry in which they previously participated. In 1.28% of observations, firms add products in a new 4-digit CIC industry. Decomposing the product addition at the sector level, in 2.93% of observations, firms add products in the 2-digit CIC sector in which they operate, while in 0.65% of observations, firms add products in the new 2-digit CIC sector. Thus, most firms tend to add products in familiar industries and sectors, indicating that product innovation, on average, is more incremental than radical innovation.

Subsequently, we estimated the effect of FDI entry on radical and incremental product innovation using the linear probability model. To this end, we use four binary dependent variables for each firm: whether a firm adds products in a 4-digit CIC industry in which they previously participated, whether a firm adds products in a new 4-digit CIC industry, whether a firm adds products in a 2-digit CIC sector in which they previously operated, and whether a firm adds products in a new 2-digit CIC sector. The results in Table 9 suggest that FDI entry mainly promotes incremental product innovation within the same industry or sector but has no significant effect on radical product innovation at the firm level.

Next, we examine the measures of product dynamics finer than product scope. In general, changes in product scope are the result of the product-switching process, in which a firm chooses to add new products and keep or drop existing products. Following the definition of Bernard et al. (2010), we compute three measures of product switching, the rate of addition (add_rate), the rate of retirement ($retire_rate$), and the net rate of change (net_rate), to capture the dynamics behind product scope. Specifically, they are defined as follows:

$$\begin{aligned} add_rate_{f,t} &= \frac{product_addition_{f,t}}{total_product_{f,t-1}} \\ retire_rate_{f,t} &= \frac{product_retirement_{f,t}}{total_product_{f,t-1}} \\ net_rate_{f,t} &= add_rate_{f,t} - retire_rate_{f,t}, \end{aligned}$$

Table 8
Heterogeneity Analysis of Entrants, Incumbents, and Exit Firms

Dependent Variable: lnScope	(1)	(2)	(3)
treat×post	0.07* (0.04)		0.10*** (0.03)
treat×post×enter	-0.02 (0.02)	0.06** (0.03)	
treat×post×incumbent		0.07* (0.04)	
treat×post×exit	-0.04 (0.03)	0.04 (0.02)	
treat×enter	0.02 (0.02)	0.02 (0.02)	
post×enter	0.01 (0.01)	0.01 (0.01)	
treat×exit	0.03** (0.02)	0.03** (0.02)	
post×exit	0.01 (0.01)	0.01 (0.01)	
enter	-0.02*** (0.01)	-0.02*** (0.01)	
exit	-0.01 (0.01)	-0.01 (0.01)	
Dependent Variable's Mean	0.26	0.26	0.33
Firm fixed effects	N	N	Y
Industry fixed effects	Y	Y	N
Year fixed effects	Y	Y	Y
Pre-determined industry controls	Y	Y	Y
Time-varying firm controls	Y	Y	Y
Tariff in 2001×year dummy	Y	Y	Y
Proportion of SOE in 2001 ×year dummy	Y	Y	Y
Constant	-0.23** (0.09)	-0.23** (0.09)	0.31*** (0.10)
Observations	257,473	257,473	58,159
R-squared	0.43	0.43	0.84

Note: 1) The Column (3) contains firms present throughout the whole sample period (stayers).

2) The variable *treat* is the treatment variable associated with the 2002 policy change. The time-varying firm control variables are log of firm output, log of firm age, log of firm's capital intensity, firm's financial ability, indicator for exporter, and indicator for SOE. Tariff is the 4-digit industry-level ad volorem tariff in 2001.

Proportion of SOE is calculated by the ratio of SOE in each 4-digit industry in 2001. 3) ***, **, * denote significance at the 1%, 5% and 10% level, respectively. The standard errors reported in parentheses are clustered at the level 4-digit industry.

Table 9
Regressions of Radical and Incremental Innovation

Dependent Variables:	Add same industry (1)	Add different industry (2)	Add same sector (3)	Add different sector (4)
treat×post	0.01* (0.01)	0.001 (0.002)	0.02* (0.01)	-0.0002 (0.002)
Dependent Variable's Mean	0.03	0.02	0.04	0.01
Firm fixed effects	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y
Pre-determined industry controls	Y	Y	Y	Y
Time-varying firm controls	Y	Y	Y	Y
Tariff in 2001×year dummy	Y	Y	Y	Y
Proportion of SOE in 2001 × year dummy	Y	Y	Y	Y
Constant	-0.06* (0.03)	0.02 (0.02)	-0.05 (0.03)	0.01 (0.01)
Observations	234,223	234,223	234,223	234,223
R-squared	0.37	0.30	0.37	0.29

Note: 1) The dependent variables are indicators for adding a new product in the same industry, a new industry, the same sector, and a new sector, respectively.

2) The variable *treat* is the treatment variable associated with the 2002 policy change. The time-varying firm control variables are log of firm output, log of firm age, log of firm's capital intensity, firm's financial ability, indicator for exporter, and indicator for SOE. Tariff is the 4-digit industry-level ad volorem tariff in 2001. Proportion of SOE is calculated by the ratio of SOE in each 4-digit industry in 2001.

3) ***, **, * denote significance at the 1%, 5% and 10% level, respectively. The standard errors reported in parentheses are clustered at the level 4-digit industry.

Table 10
Regressions of Product Turnover Rates

Dependent Variables:	Add rate (1)	Retire rate (2)	Net rate (3)
treat×post	0.02** (0.01)	-0.002 (0.01)	0.02*** (0.01)
Dependent Variable's Mean	0.04	0.03	0.01
Firm fixed effects	Y	Y	Y
Year fixed effects	Y	Y	Y
Pre-determined industry controls	Y	Y	Y
Time-varying firm controls	Y	Y	Y
Tariff in 2001×year dummy	Y	Y	Y
Proportion of SOE in 2001×year dummy	Y	Y	Y
Constant	-0.01 (0.04)	0.08*** (0.02)	-0.09*** (0.03)
Observations	234,223	234,223	234,223
R-squared	0.35	0.36	0.28

Note: 1) The dependent variables are the rate of adding new products, the rate of retiring products, and the net rate of adding new products, respectively.

2) The variable *treat* is the treatment variable associated with the 2002 policy change. The time-varying firm control variables are log of firm output, log of firm age, log of firm's capital intensity, firm's financial ability, indicator for exporter, and indicator for SOE. Tariff is the 4-digit industry-level ad volorem tariff in 2001. Proportion of SOE is calculated by the ratio of SOE in each 4-digit industry in 2001.

3) ***, **, * denote significance at the 1%, 5% and 10% level, respectively. The standard errors reported in parentheses are clustered at the level 4-digit industry.

where *product_addition* is the number of products added, *product_retirement* is the number of retired products, and *total_product* is the total number of products. Using the three new measures as dependent variables, we rerun our regressions and report the results in columns (1)–(3) of Table 10. The results suggest that relaxation in FDI regulation is associated with a significant increase in the rate of adding new products (*add_rate*) but has no effect on the rate of dropping existing products (*retire_rate*). The effect of FDI on the net rate of product addition (*net_rate*), estimated to be 0.02, is mostly accounted for by the effect of FDI on the addition rate (0.02).

5.3. Accounting for effects of trade liberalization, yuan appreciation, and reduction in trade policy uncertainty

During our sample period of 2000 to 2007, trade liberalization, the appreciation of the Chinese yuan, and reduction in trade policy uncertainty (TPU) may also have contributed to the increase in the product scope of Chinese firms. Trade liberalization affects product scope through price and cost channels. First, after trade liberalization lowers market prices and markups, firms are forced to give up products with low margins (Baldwin and Gu, 2009; Bernard et al., 2011; Eckel and Neary, 2010; Mayer et al., 2014). Second, as marginal cost increases with product scope, firms choose to reduce product scope and, hence, reduce the marginal cost to meet the competition associated with trade liberalization (Nocke and Yeaple, 2014).

First, to account for trade liberalization at both the industry and firm levels in the regression, we follow the approach of Yu (2015). For industry-level trade liberalization, we introduce 2-digit industry-year fixed effects. For firm-level differences in tariffs on imported inputs, the measure is constructed as follows:

$$\text{input tariff}_{f,t} = \sum_{k \in O} \frac{m_{f,t0}^k}{\sum_{k \in M} m_{f,t0}^k} \cdot \tau_t^k \quad (3)$$

where k, f, t and $t0$ are the indices for product, firm, time, and initial period, respectively. $\text{input tariff}_{f,t}$ is the average tariff level on firm f 's imported inputs in year t , $m_{f,t0}^k$ is firm f 's import of product k in the initial period $t0$, and τ_t^k is the tariff level for product k in year t . M denotes the set of products imported by firm f . As the processing trade is exempt from import tariffs, we use information on ordinary trade to compute $\text{input tariff}_{f,t}$. To prevent the measure of the average tariff from being affected by the endogenous response in import value to tariffs, we use only the initial import value as the weight.

As for the measure of the tariff on output, Yu (2015) pointed out that its construction should use the fraction of domestic sales in the total output. Unfortunately, this information was unavailable. We follow Melitz (2003) and posit that an efficient firm should sell to both domestic and foreign markets. Thus, sales to a foreign market imply sales in the domestic market. Subsequently, we follow the practice of Yu (2015) to assume that the fraction of domestic sales is the same as foreign sales and replace the former with the latter. Consequently, the output tariff of firm f in year t can be expressed as

$$\text{output tariff}_{f,t} = \sum_k \frac{X_{f,t0}^k}{\sum_k X_{f,t0}^k} \cdot \tau_t^k \quad (4)$$

Table 11
Regressions with Trade Liberalization and Intermediate Inputs

Dependent Variable: Inscope	(1)	(2)	(3)	(4)	(5)	(6)
treat×post	0.03** (0.01)	0.04** (0.02)	0.05*** (0.02)	0.03** (0.01)	0.01** (0.01)	0.01** (0.01)
output tariff	0.003 (0.01)					
input tariff		0.02 (0.01)				
ln(no. of intermediate inputs)			0.002 (0.003)			
TPU×post				-0.21*** (0.06)	0.06 (0.13)	0.005 (0.01)
Dependent Variable's Mean	0.35	0.38	0.26	0.26	0.26	0.26
Firm fixed effects	Y	Y	Y	Y	Y	Y
2-digit industry-year fixed effects	Y	Y	N	N	N	N
Year fixed effects	N	N	Y	Y	Y	Y
Pre-determined industry controls	Y	Y	Y	Y	Y	Y
Time-varying firm controls	Y	Y	Y	Y	Y	Y
Tariff in 2001×year dummy	Y	Y	Y	Y	Y	Y
Proportion of SOE in 2001 × year dummy	Y	Y	Y	Y	Y	Y
Constant	0.09 (0.08)	0.06 (0.11)	0.21** (0.08)	0.26*** (0.09)	0.01 (0.12)	0.05 (0.03)
Observations	37,834	20,315	338,201	338,201	297,293	297,293
R-squared	0.93	0.92	0.89	0.89	0.91	0.91

Note: 1) The variable *treat* is the treatment variable associated with the 2002 policy change. *output tariff*, *input tariff* and *ln(no. of intermediate inputs)* are the trade-weighted output tariff, trade-weighted input tariff, and the log of the number of intermediate inputs used at the firm level, respectively. The time-varying firm control variables are log of firm output, log of firm age, log of firm's capital intensity, firm's financial ability, indicator for exporter, and indicator for SOE. Tariff is the 4-digit industry-level ad valorem tariff in 2001. Proportion of SOE is calculated by the ratio of SOE in each 4-digit industry in 2001.

2) ***, **, * denote significance at the 1%, 5% and 10% level, respectively. The standard errors reported in parentheses are clustered at the level 4-digit industry.

where $X_{f,t0}^k$ is the export of product k by firm f in the initial period t_0 , and τ_t^k is the export tariff. Similar to the construction of *input tariff* $f_{f,t}$, we use the initial export value as the weight throughout all years.

In the first two columns of Table 11, we introduce the average tariffs on imported inputs and outputs, respectively. All regressions include 2-digit industry-year fixed effects. The coefficient of the treatment variable remained positive and statistically significant. Overall, after accounting for industry- and firm-level trade liberalization, we find that FDI entry has positive effects on domestic firms' product scope.

Second, as imported intermediate goods often embody better technology (Blalock and Veloso, 2007), and the import of intermediate inputs can boost the product scope of domestic firms (Kee, 2015), the yuan appreciation starting from July 2005 can increase product scope by making imported intermediate goods cheaper. Thus, we include the log of the number of imported intermediate goods in the regression to verify that FDI spillover effects are not confounded by the import of intermediate inputs.

We apply the Broad Economic Categories (BEC) classification to identify imported intermediate goods from a firm's imports and include the log of the number of intermediate products plus one in the benchmark regression. To be precise, we regard products with BEC codes of '111', '121', '21', '22', '31', '322', '42' and '53' as intermediate goods.¹³ For non-importer firms, the number of imported intermediate goods was set to 0. The results, as shown in the last column of Table 11, indicate that the effect of FDI on product scope remains positive and significant, whereas imported intermediate inputs do not appear to have a significant effect.

Third, around the time of relaxation in FDI regulations, China's entry into WTO may also affect product scope by reducing TPU. The trade literature typically measures reduction in TPU in China by the difference between Column 2 tariff rates¹⁴ and Most Favored Nations (MFN) tariff rates. To obtain industry-specific measure of TPU, we first compute the TPU at product level and then aggregate it to the industry level. We follow three typical methods to compute the TPU at the product level.

As for the first measure, we follow Pierce and Schott (2016) to calculate product-level TPU as

$$TPU_{h,2001}^1 = \tau_{h,2001}^{col\ 2} - \tau_{h,2001}^{mfn} \quad (5)$$

where $\tau_{h,2001}^{col\ 2}$ is the Column 2 ad valorem tariff rate applied to product h (defined at HS 6-digit level) in 2001, and $\tau_{h,2001}^{mfn}$ the MFN tariff rate of the same product in 2001. The larger of the difference, the greater is the reduction in TPU post the granting of Permanent Normal Trade Relations (PNTR) to China by the US in 2002.

¹³ These BEC codes correspond to food and beverages-primary-mainly for industry, food and beverages-processed-mainly for industry, industrial supplies not elsewhere specified-primary, industrial supplies not elsewhere specified-processed, fuels and lubricants-primary, fuels and lubricants-processed-other, capital goods (except transport equipment) and parts and accessories thereof-parts and accessories, and transport equipment and parts and accessories thereof-parts and accessories.

¹⁴ Column 2 tariff rates are the duty rates applied to countries with which the United States do not have "Normal Trade Relations".

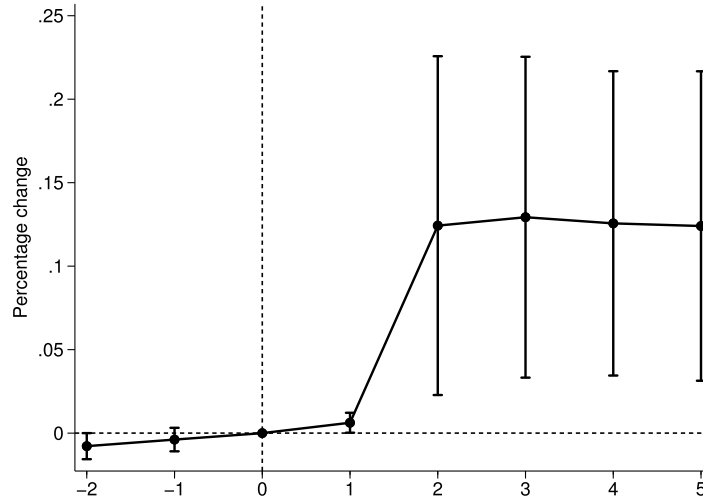


Fig. 4. Effects of FDI Policy Change on Product Scope Based on Event Study

Source: authors' calculation.

To obtain the second measure, we follow Handley and Limão (2017) to define product-level TPU as

$$TPU_{h,2001}^2 = 1 - \left(\frac{\tau_{h,2001}^{col\ 2}}{\tau_{h,2001}^{mfn}} \right)^{-\sigma} \quad (6)$$

where the parameter σ is set to 3.

The third measure of TPU is an alternative measure proposed also by Handley and Limão (2017):

$$TPU_{h,2001}^3 = \ln \left(\frac{\tau_{h,2001}^{col\ 2}}{\tau_{h,2001}^{mfn}} \right). \quad (7)$$

After obtaining the product-level TPU, we map the HS 6-digit products into CIC 4-digit industry classification of China, and compute the industry level trade policy uncertainty, TPU_i , as the simple average of TPU of 6-digit products in industry i . Subsequently, we modified the benchmark regression as

$$\ln(scope_{fit}) = \beta_0 + \beta_1 treat_i \times post_t + \beta_2 TPU_i \times post_t + \rho \cdot X_{fit} + \gamma_f + \delta_t + \epsilon_{fit} \quad (8)$$

Regressions based on the three measures of TPU defined in equations (5) to (7) are reported in Columns (4) to (6) of Table 11. The results indicate that after controlling for the effect of reduction in trade policy uncertainty, FDI positively affects product scope of Chinese firms. As for the effect of TPU reduction post China's entry of WTO and gaining PNTR status from the US, we find that reduction in TPU as defined in Pierce and Schott (2016) is associated with a reduction in product scope which is the opposite to the effect of FDI. This may be because that reduction in TPU encourages further specialization by Chinese firms. When the other TPU measures are used, the estimated effects are insignificant.

5.4. Evidence from event study

We follow Kline (2012) and Abebe et al. (2022) and employ the event study method to verify the parallel trend assumption. In equation (9), we introduce a set of interactions between the treatment indicator and year dummies ($treat_i \times post_t$) ^{j} to capture the time trend.

$$\ln(scope_{fit}) = \beta_0 + \sum_{j=-2, j \neq 0}^{j=5} \gamma_j (treat_i \times post_t)^j + \rho X_{fit} + \gamma_f + \delta_t + \epsilon_{fit}. \quad (9)$$

The dummy variable $post_t^j$ takes the value of 1 if year t is j years apart from the impact year and 0 otherwise. When negative, the superscript j indicates that year t is j years before the relaxation of the FDI entry policy. By symmetry, a positive value of j indicates that year t is j years after the relaxation of FDI entry policy. After centering the estimates on the year of the policy change (year 0), we summarize the trend of the coefficient on ($treat_i \times post_t$) ^{j} with their respective 95% confidence intervals in Fig. 4.

Clearly, when j is equal to -2 and -1, the coefficients on the dummies are not statistically different from 0. Two years after the policy change—for the j value of 2—the coefficients of the interactions increase sharply and become statistically significant, indicating a strong effect of FDI entry on the product scope. The effect of policy change emerges with a lag, which might appear puzzling. A natural conjecture is that the lag in policy effect is caused by the delayed increase in actual FDI entry post-policy change

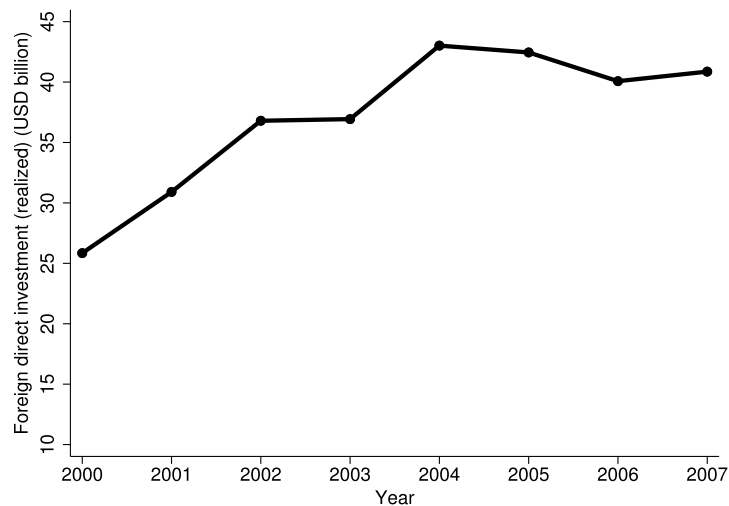


Fig. 5. Foreign Direct Investment Flows in the Manufacturing Sector in China

Source: China Statistical Yearbook.

in 2002. To this end, we plot the inflows of FDI in manufacturing against time in Fig. 5 which shows that FDI inflows indeed exhibited notable increase in 2004. In subsequent years, the coefficients on the interaction terms are of similar magnitude, which suggests that the effect of FDI entry on product scope is persistent.

5.5. Within-firm effect of FDI on product scope

To complement our analysis on spillover effects of FDI on domestic firms, in this subsection we analyze the direct effects of FDI, i.e. the within-firm effect of increase in foreign investment on product scope of firms with foreign ownership. We present evidence from three different regression designs. In the first, we add the fraction of foreign capital in total paid-in capital ($foreign\ share_{ft}$) to a regression of log product scope on the same set of controls used in the benchmark regression. Results reported in Column (1) of Table 12 indicate that a 1% rise in foreign ownership is associated with 0.0059 increase in log product scale, an effect that is statistically and practically significant.

Second, we adopt an event study framework to compare product scope of the group firms that experienced increase in foreign ownership (denoted by the label *increase* in regression) relative to the rest. In Column (2) of Table 12, the group is defined as firms which report positive change in foreign ownership around 2002. The regression in Column (3) clearly demonstrates that firms reporting a positive increase in foreign ownership after 2002 subsequently expanded their product scope more than other firms. In an unreported regression, we compare the change in product scope for the group of firms with above the mean level of change in foreign ownership to the rest, similar results are obtained.

Third, we follow Brambilla (2009) to introduce indicators for SOEs, minority foreign-owned firms, majority foreign-owned firms, and fully foreign-owned firms to the regression of product scope. Domestic private firms are the reference group. In Column (3) of Table 12, we can see that minority foreign-owned and fully foreign-owned firms report large product scope than domestic private firms.

Overall, results from the three regression designs all suggest that an increase in foreign ownership in a firm is associated with an expansion of product scope in the same firm. In other words, the direct effects of FDI on product scope are likely to be positive. Meanwhile, we recognize that the identification of direct effects in some of the regressions is tenuous and hence the results should be interpreted with caution.

6. Conclusion

During the era of ‘reform and opening up’, China has emerged as a significant economy in terms of attracting FDI and contributing to global manufacturing output. This study aims to investigate whether and how FDI contributes to product innovation in China. Leveraging the WTO entry-related policy change in 2002 that eased FDI entry for several industries, we used the DID model to estimate the causal effect of FDI entry on domestic Chinese firms’ product scope.

Our analysis of product scope yields four main findings. Firstly, FDI is an important factor in explaining product innovation in China. Following the WTO-related policy overhaul on FDI regulation in 2002, the product scope of domestic Chinese firms in industries that enjoyed a relaxation in FDI entry increased by 5% on average. Secondly, apart from the horizontal spillover effects observed within industries, FDI also influences product scope through vertical linkages. The entry of FDI in upstream industries positively impacts the product scope of domestic firms, while FDI entry in downstream industries has a negative effect on their product scope. Thirdly, we identify two channels through which FDI contributes to product innovation of domestic firms. FDI stimulates domestic firms to increase R&D expenditures and shortens the technological distance between Chinese industries and the

Table 12
Results for firms' FDI Entry and Product Scope

Dependent Variable: lnScope	(1)	(2)
For Share	0.01* (0.003)	
Minority (10-49%)		0.01** (0.004)
Majority (50-99%)		-0.003 (0.004)
Fully (100%)		0.01*** (0.003)
Dependent Variable's Mean	0.28	0.28
Firm fixed effects	Y	Y
Year fixed effects	Y	Y
Pre-determined industry controls	Y	Y
Time-varying firm controls	Y	Y
Tariff in 2001×year dummy	Y	Y
Proportion of SOE in 2001×year dummy	Y	Y
Constant	0.18*** (0.01)	0.18*** (0.01)
Observations	522,092	525,720
R-squared	0.88	0.88

Notes: 1)

2) The time-varying firm control variables are log of firm output, log of firm age, log of firm's capital intensity, firm's financial ability, indicator for exporter, and indicator for SOE. Tariff is the 4-digit industry-level ad volorem tariff in 2001. Proportion of SOE is calculated by the ratio of SOE in each 4-digit industry in 2001.

3) ***, **, * denote significance at the 1%, 5% and 10% level, respectively. The standard errors reported in parentheses are clustered at the level firm.

world frontier. Lastly, while there is a positive effect of FDI on product scope, it does have limitations. Introduction of new products is more likely to occur within the same industry rather than across industries, suggesting that FDI primarily promotes incremental innovation. The main findings hold robust when we include alternative factors that may contribute to product innovation and adopt alternative measures of product innovation.

Overall, based on Chinese data, our study presents compelling evidence that entry of FDI contributes significantly to product innovation of domestic firms. This finding carries important implications for economic growth and development. However, much remains to be explored in research on product innovation. Further research in the context of different countries is needed to understand how FDI affects innovation in developing countries as a whole. Moreover, it is important to note that our identification and estimation rely on a one-time policy change. Future research could benefit from improved data sources and alternative identification strategies to examine the long-term effects of FDI on product innovation.

Declaration of competing interest

The author declares that she has no relevant or material financial interests that relate to the research described in this paper.

Data availability

The authors do not have permission to share data.

Appendix A

In this appendix, we justify the definition of products in the China Product Output Database. We follow the literature (e.g., Bernard et al. 2010) in identifying products with the 5-digit product codes. In this database, a total of 729 products are coded in 5-digit codes (hereinafter, “original codes”) which are augmented by product names. Because the 5-digit original codes we used are not organized in the SIC system, they are not equivalent to the 5-digit SIC codes used by Bernard et al. (2010). To ensure that the original codes can be plausibly interpreted as products, we map the original codes onto the 5-digit product codes in the Central Product Classification (CPC version 1.0) published by the United Nations to show the comparability between product definitions.

We make the mapping from the original codes to 5-digit CPC codes in two steps. In the first step, we map the original codes to the codes in the Product Code Classification for Statistics¹⁵, published by the National Bureau of Statistics of China (NBSC) in 2010. The latter codes are the standard product codes used by NBSC since 2010. In the second step, we use the concordance published by the

¹⁵ The Chinese name is *Tongji Yong Chanpin Fenlei Daima*.

Table A1
OECD Technology Classification (GB/T 4754-2002)

High-tech industries	
Manufacture of medicines	27
Manufacture of medical equipment and instruments	368
Manufacture of postal machinery and equipment	3693
Manufacture of aerospace vehicles	376
Manufacture of power electronics components	3924
Manufacture of communications equipment, computers and other electronic equipment (excl. other electronic equipments)	40 (excl. 4090)
Measuring instruments and machinery for cultural activity and office work (excl. special instruments for teaching)	41 (excl. 4126)
Medium-tech industries	
Processing of salt	1493
Manufacture of stationery	2411
Manufacture of ink	2414
Processing of petroleum, coking, and nuclear fuel	25
Manufacture of chemical raw materials and chemical products	26
Manufacture of chemical fibers (excl. chemical fiber pulp)	28 (excl. 2811)
Manufacture of rubber (excl. rubber boots)	29 (excl. 2960)
Manufacture of plastics (excl. plastic shoes)	30 (excl. 3081)
Manufacture of non-metallic mineral products (excl. waterproof building materials)	31 (excl. 3134)
Smelting and processing of ferrous metals	32
Smelting and processing of non-ferrous metals	33
Manufacture of metal products (excl. coins and precious metal laboratory supplies)	34 (excl. 3491)
Manufacture of general purpose machinery	35
Manufacture of special equipment (excl. manufacture of medical equipment and instruments, and postal machinery and equipment)	36 (excl. 368; 3693)
Manufacture of transport equipment (excl. aerospace vehicles)	37 (excl. 376)
Manufacture of electrical machinery and equipment (excl. power electronics components)	39 (excl. 3924)
Manufacture of other electronic equipments	4090
Manufacture of sculpturing handicraft	4211
Manufacture of metal handicraft	4212
processing of mirror and similar products making	4221
Low-tech industries	
Processing of food from agricultural products	13
Manufacture of foods (excl. processing of salt)	14 (excl. 1493)
Manufacture of beverages	15
Manufacture of tobacco	16
Manufacture of textiles	17
Manufacture of textile, apparel, footwear, and caps	18
Manufacture of leather, fur, feather and related products	19
Processing of timber, manufacture of wood, bamboo, rattan, palm, and straw products	20
Manufacture of furniture	21
Manufacture of paper and paper products	22
Printing and recorded media	23
Manufacture of articles for culture, education and sport activity (excl. manufacture of ink)	24 (excl. 2414)
Manufacture of other daily chemicals	2679
Manufacture of sanitary materials and medical supplies	2770
Manufacture of chemical fiber pulp	2811
Manufacture of rubber boots	2960
Manufacture of plastic shoes	3081
Manufacture of waterproof building materials	3134
Manufacture of coins and precious metal laboratory supplies	3491
Manufacture of special instruments for teaching	4126
Manufacture of artwork and other manufacturing (excl. sculpturing handicraft; metal handicraft; mirror and similar products making; coal products)	42 (excl. 4211; 4212; 4221; 4230)

Note: This table reports the detailed technology intensity industry classification based on OECD. Different from the CIC code adjusted based on Brandt et al. (2012) in the empirical regressions, in order to show the industry name and better understand the classification, we show this table with the CIC system of GB/T 4754-2002 version.

NBSC to link the codes in the Product Code Classification to product codes in the CPC (version 1.0) published by the United Nations. In the end, we identified 612 products by CPC codes. By mapping 727 products by original codes to 612 products by CPC codes, we show that the original codes can be used to define products in the sample.

In principle, we can map the original codes to SIC codes in two additional steps, that is, by using the concordance provided by the United Nations to map the codes in the CPC version 1.0 to 6-digit HS codes, then map the HS codes to SIC codes with the standard concordance. However, it is most unfortunate that the codes in CPC version 1.0 are 5-digit codes and hence much coarser than the 6-digit HS codes. This means that a particular product code in CPC version 1.0 can map onto multiple (on average 12) 6-digit HS codes based on the concordance of CPC 1.0 and HS96 from WITS. We chose not to take this route because the additional mapping would systematically inflate the number of products in the sample.

Table A2
Classification Based on 2002 Input-Output Table of China

Upstream	Manufacture of basic chemical raw materials
	Smelting of non-ferrous metals
	Waste products
	Manufacture of special chemical products
	Manufacture of coatings, pigments, inks and similar products
	Manufacture of Electronic components
	Smelting of ferroalloy
	Manufacture of synthetic materials
	Coking
	Smelting of iron
	Manufacture of chemical fibers
	Processing of non-ferrous metals
	Manufacture of cultural and office machinery
	Manufacture of plastic products
	Smelting of Steel
	Processing of petroleum and nuclear fuel
	Manufacture of other computer equipment
	Manufacture of glass and glass products
	Manufacture of paper and paper products
	Manufacture of rubber products
	Manufacture of other electrical machinery and equipment
	Manufacture of auto parts and accessories
	Manufacture of shipbuilding and floating device
	Manufacture of pesticide
Midstream	Manufacture of leather, fur, feather and related products
	Manufacture of computer machine
	Manufacture of daily chemical products
	Manufacture of metal processing machinery
	Processing of feed
	Manufacture of home audio visual equipment
	Manufacture of boiler and prime mover
	Manufacture of other non-ferrous metals products
	Manufacture of knitwear, knitted goods and their products
	Hemp textile, silk textile and finishing
	Manufacture of refractory products
	Manufacture of articles for culture
	Processing of wood and manufacture of wood, bamboo, rattan, palm and grass products
	Manufacture of motor
	Manufacture of fertilizer
	Processing of steel
	Manufacture of other communication and electronic equipment
	Printing and reproduction of recording media
	Manufacture of other general purpose machinery
	Manufacture of textile products
	Wool textile, dyeing and finishing
Downstream	Manufacture of metal products
	Manufacture of measuring instruments
	Cotton, chemical fiber textile, printing, dyeing and finishing
	Processing of other food and manufacture of food
	Manufacture of toy, sports and entertainment products
	Manufacture of Special purpose machinery for agriculture, forestry, animal husbandry and fishery
	Manufacture of crafts and art production
	Manufacture of tobacco products
	Manufacture of other beverage
	Milling of grain
	Sugar
	Processing of vegetable oil
	Processing of aquatic products
	Manufacture of textile, apparel, footwear, and caps
	Manufacture of medicines
	Manufacture of alcohol and beverage
	Slaughtering and processing of meat
	Manufacture of household appliance
	Manufacture of cement, lime and gypsum
	Manufacture of other special purpose machinery
	Manufacture of communication equipment
	Manufacture of automobile
	Manufacture of other transport equipment
	Manufacture of furniture
	Manufacture of ceramics
	Manufacture of railway transport equipment

Table A3
Regressions of Instrumental Variables

IV	(1)	(2)	(3)
Panel A: First-stage estimation (dependent variable: FDI_ind_share).			
$treat \times post$	0.02*** (0.01)	0.02*** (0.01)	0.02*** (0.01)
Panel B: First-stage estimation (dependent variable: $lnScope$).			
FDI_ind_share	3.51** (1.68)	3.45** (1.67)	3.12** (1.47)
Panel C. Weak instrument test			
Cragg-Donald Wald F statistic	1616.254 [16.38]	1628.598 [16.38]	1394.239 [16.38]
Firm fixed effects	Y	Y	Y
Year fixed effects	Y	Y	Y
Pre-determined industry controls	Y	Y	Y
Time-varying firm controls	N	Y	Y
Tariff in 2001×year dummy	N	N	Y
Proportion of SOE in 2001×year dummy	N	N	Y
Observations	338,201	338,201	338,201

Notes: 1) The time-varying firm control variables are log of firm output, log of firm age, log of firm's capital intensity, firm's financial ability, indicator for exporter, and indicator for SOEs. Tariff is the 4-digit industry-level ad valorem tariff in 2001. Proportion of SOE is calculated by the ratio of SOE in each 4-digit industry in 2001.

2) ***, **, * denote significance at the 1%, 5% and 10% level, respectively. The standard errors reported in parentheses are clustered at the level 4-digit industry-year.

Appendix B

Our benchmark regression relies on the DID framework for identification. In this Appendix, we present results based on two alternative identification strategies, instrumental variable (IV) regressions and synthetic DID (see Table A3). We follow Lu et al. (2017) in using the revision of the Catalog of Industries for Guiding Foreign Investment (hereinafter the Catalog of Industries) in 2002 as an instrumental variable for the share of foreign firms in the industry. In the first stage, we regress the share of foreign firms on policy changes at industry level, then regress product scope on FDI share in the second stage.

The first stage involves a difference-in-difference (DID) regression of the share of foreign firms in a given industry (FDI_ind_share) on a policy change that affected FDI entry in some industries in 2002. The regression equation is

$$FDI_ind_share_{it} = \beta_0 + \beta_1 treat_i \times post_t + \rho X_{fit} + \gamma_f + \delta_t + \epsilon_{fit}, \quad (B.1)$$

where f , i , and t are indicators of the firm, 4-digit industry code, and year, respectively. Following Aitken and Harrison (1999) and Javorcik (2004), $FDI_ind_share_{it}$ is the share of foreign firms in the industry i to which firm f belongs. The degree of FDI entry is measured as

$$FDI_ind_share_{it} = \frac{\sum_{f \in i} FDI_firm_{fit} \times value_added_{fit}}{\sum_{f \in i} value_added_{fit}}, \quad (B.2)$$

where FDI_firm_{fit} is the share of foreign capital in total paid-in capital of the firm and $value_added_{fit}$ is the value added of the firm.

The variable $treat_i$ is a dummy variable that equals 1 if industry i becomes more open to FDI entry in 2002, and 0 if the industry experienced no change in entry policy. Thus, the group of industries that experienced no policy change is the reference group. The variable $post_t$ is an indicator that takes the value of 1 for years after 2002, and 0 otherwise. The quantities γ_f , and δ_t are firm fixed effects, and time fixed effects, respectively. The error term is ϵ_{fit} .

The second-stage regression estimates the causal effect of FDI entry on product scope, and the regression equation is specified as

$$lnscope_{fit} = \beta_0 + \beta_1 FDI_ind_share_{it} + \rho X_{fit} + \gamma_f + \delta_t + \epsilon_{fit}, \quad (B.3)$$

where the variable $lnscope_{fit}$ is the log of the product scope of a firm, and product scope is measured as the number of products produced by firm f at year t in industry i .

Our control variables (X_{fit}) are firm size, measured as log output ($lnOutput$); log firm age ($lnAge$); capital intensity, measured as log capital to labor ratio (lnK/L); financing ability ($FinAbility$); an indicator for state-owned enterprises (SOE); an indicator for exporting activities ($export$). The error term was ϵ_{fit} . We clustered the standard error at 4-digit industry-year level. The error term is ϵ_{fit} .

Second, we use the synthetic control DID method of Arkhangelsky et al. (2021) to identify the effect of FDI on product scope. Relative to conventional DID, synthetic DID provides a more robust identification strategy because the use of a synthetic control group reduces reliance on the parallel trend assumption. The regression results reported in the first column of Table A4 confirm that FDI has a significant and positive effect on product scope. As synthetic methods require a balanced panel, the sample size is 49,432, which is significant below 338,201 in the baseline regression. While the size of the coefficient (0.17) is large in a practical

Table A4
Alternative Identification Using Synthetic DID

Dependent Variable: lnScope	Synthetic DID (1)	DID (2)
treat×post	0.17*** (0.01)	0.10*** (0.03)
Dependent Variable's Mean	0.31	0.31
Firm fixed effects	Y	Y
Year fixed effects	Y	Y
Pre-determined industry controls	Y	Y
Time-varying firm controls	Y	Y
Tariff, 2001×year dummy	Y	Y
Proportion of SOE, 2001×year dummy	Y	Y
Constant	-	0.32*** (0.09)
Observations	49,432	49,432

Notes: 1) The variable *treat* is the treatment variable associated with the 2002 policy change. The time-varying firm control variables are log of firm output, log of firm age, log of firm's capital intensity, firm's financial ability, indicator for exporter, and indicator for SOE. Tariff is the 4-digit industry-level ad valorem tariff in 2001. Proportion of SOE is calculated by the ratio of SOE in each 4-digit industry in 2001.

2) ***, **, * denote significance at the 1%, 5% and 10% level, respectively. The standard errors reported in parentheses are clustered at the level 4-digit industry.

sense in the balanced panel, the difference from the baseline estimate (0.05) cannot be attributed to the difference in identification strategy alone. This is because the baseline results are obtained from an unbalanced sample. To see the difference in estimated effects between the baseline DID and the synthetic DID, in column (2) of Table A4, we report the baseline DID regression estimated with the balanced panel. The coefficient of the baseline DID (0.10) was significantly smaller than that of the synthetic DID (0.17). Thus, the estimated spillovers are robust and large for the synthetic DID model.

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